Environmental Studies

Assignment2

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01 SITTING POSTURE ERGONOMIC ANALYSIS

1.1 Personal Anthropometric Data

Measurement	Value	Classification
Height (A)	155 cm	25th percentile female
Weight	64 kg	Healthy range
BMI	26.6	Overweight (25-30)

1.2 Key Dimensional Values from Lecture Diagrams

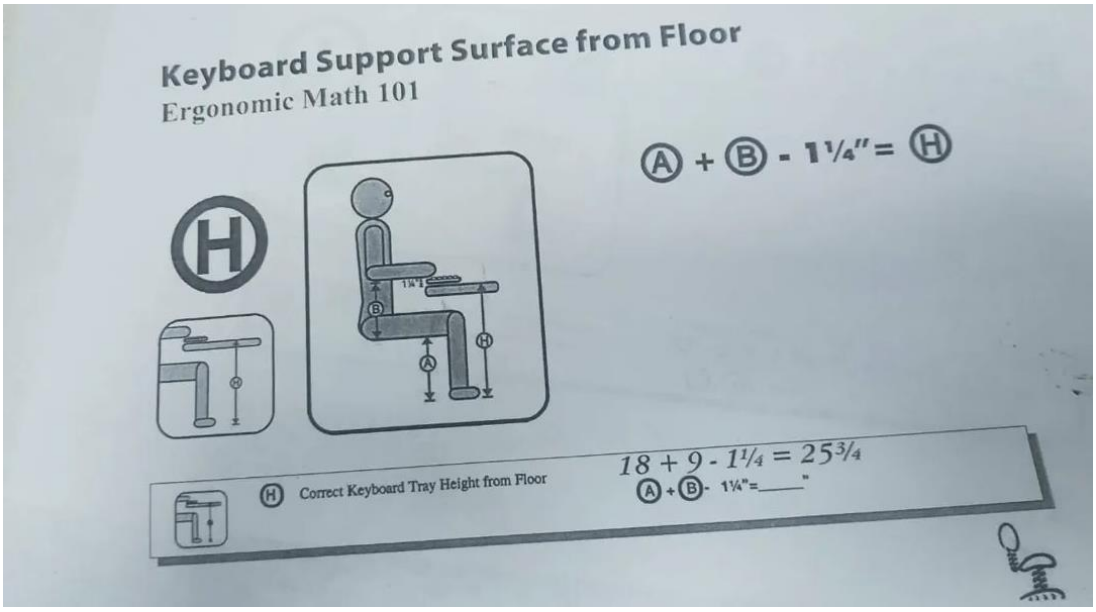


Figure 1.1: Keyboard Support Surface from Floor — Ergonomic Math 101

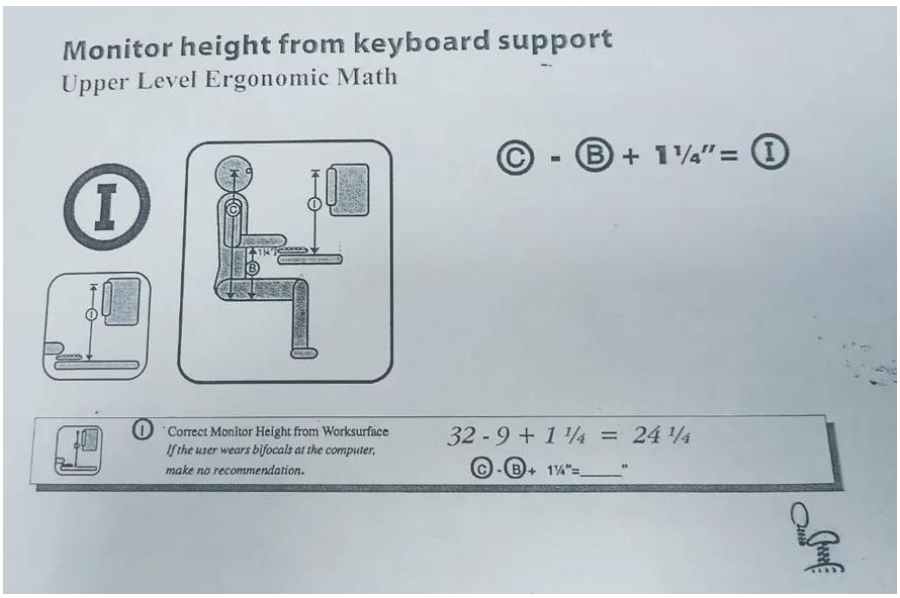


Figure 1.2: Monitor Height from Keyboard Support — Upper Level Ergonomic Math

From the lecture diagrams and applying my dimensions and the seat in the college , the following ergonomic equations are provided:

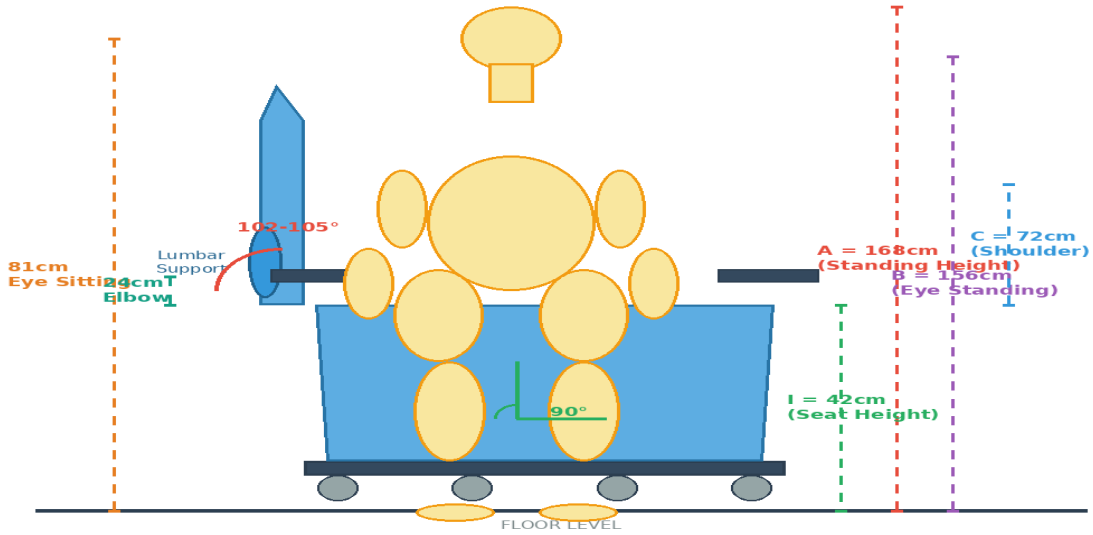
```
H = A + B - 1¼″
I = C - B + 1¼″
```

For the subject (Height A = 155 cm = 61.0″):

Variable	Description	Value (cm)	Value (inches)
A	Popliteal Height (seat)	42	15.2"
B	Elbow Rest Height (from seat)	24	8.7"
C	Eye Height Sitting	81	29.5"
H	Keyboard Tray from Floor	57.3	A+B-1.25" = 22.6"
I	Monitor Top from Keyboard	56.2	C-B+1.25" = 22.1"

1.3 Correct vs Incorrect Sitting Postures

✓ CORRECT SITTING POSTURE

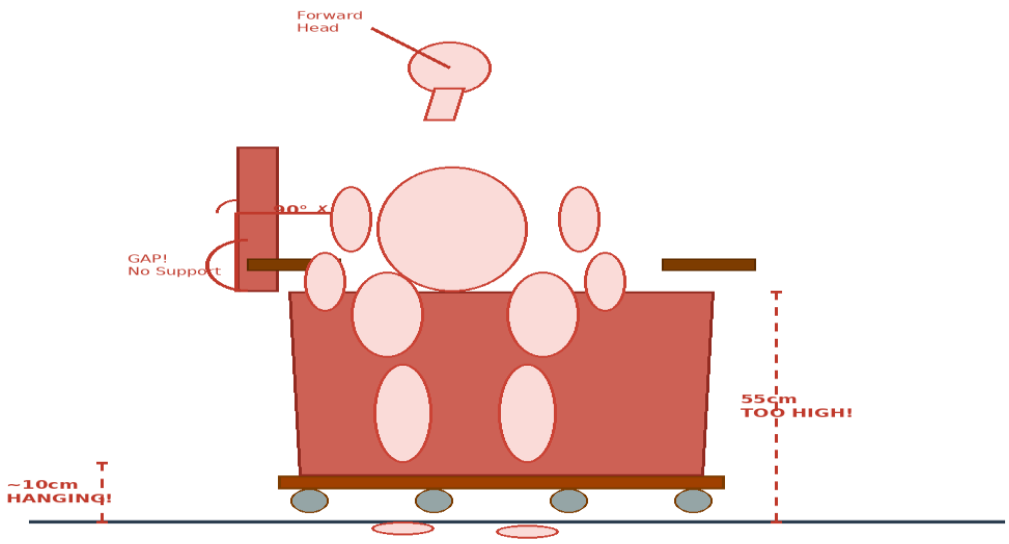


KEY ERGONOMIC FEATURES

- ✓ Backrest angle 102-105° with full lumbar support
- ✓ Seat height = Popliteal height (42 cm / 16.5")
- ✓ Feet flat on floor, knees at 90-95°
- ✓ Armrests at elbow height (24 cm from seat)
- ✓ Eye level at monitor top (81 cm from floor)

Figure 1.3: CORRECT Sitting Posture — 102-105° backrest, feet flat, lumbar support

✗ INCORRECT SITTING POSTURE



COMMON ERGONOMIC ERRORS

- ✗ Seat 13cm too high (55cm vs 42cm) → Feet hang, poor circulation
- ✗ Backrest at 90° (should be 102-105°) → Lower back pain
- ✗ No lumbar support → Chronic back strain
- ✗ Forward head posture → Neck and shoulder tension
- ✗ Armrests not used → Upper body fatigue

Figure 1.4: INCORRECT Sitting Posture — 90° backrest, feet hanging, no lumbar support

1.4 Ergonomic Standards and Deviations

Feature	Correct Posture	Incorrect Posture	Consequence
Seat Height (A)	42 cm (popliteal)	55 cm (too high)	Circulation problems
Backrest Angle	102-105°	90° (upright)	Lower back pain
Lumbar Support	Full contact	Gap present	Disc compression
Feet Position	Flat on floor	Hanging ~10cm	Poor posture
Knee Angle	90-95°	<80°	Thigh pressure

ERGONOMIC RECOMMENDATION: Use adjustable chair with seat height 38-46 cm, lumbar support, backrest recline 100-110°. Ensure feet flat on floor or use footrest.

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02 THERMAL LOAD & SOLAR RADIATION IMPACT

2.1 Problem Statement

A lecture hall (15m × 12m × 7m, 10% occupancy) houses 34 students for 2 hours. Initial temperature: 25°C. Final temperature: 32°C. Analyze thermal load from occupants and solar radiation through 6 glass windows (1m² each).

2.2 Given Data

Parameter	Value	Unit
Room dimensions (L×W×H)	15 × 12 × 7	m
Total volume	1260	m³
Occupancy rate	10%	—
Net air volume	1134	m³
Number of people	132	persons
Initial temperature (T_1)	25	°C
Final temperature (T_2)	32	°C
Temperature rise (ΔT)	7	°C
Duration	120	minutes
Windows (count × area)	6 × 1	m²
Glass transmittance	0.85	—

Solar irradiance (Cairo, noon)	700-800	W/m²
Wall U value	0.8	W/(m²·K)

2.3 Thermal Energy Calculation

Heat capacity of air:

$$Q = \rho \times V \times C_p \times \Delta T$$

Where: ρ = 1.2 kg/m³ (air density), C_p = 1005 J/(kg·°C) (specific heat)

```
Q = 1.2 × 1134 × 1005 × 7 = 9,538,000 J ≈ 9.54 MJ
```

2.4 Heat Sources

① METABOLIC HEAT (Occupants):

```
Q_people = 132 persons × 100 W/person = 13,200 W
Over 2 hours: 13,200 × 7,200 s = 95,040,000 J=95.04 MJ
```

② SOLAR RADIATION (Windows) and walls conduction:

```
A_glass = 6 windows × 1 m² = 6 m²
Q_solar = A × I × τ = 6 × 750 × 0.85 = 3,825 W
Over 2 hours: 3,825 × 7,200 s = 27.5 MJ
```

③ CONDUCTION THROUGH WALLS:

- **U** = 0.8 W/(m²·K) (Overall heat transfer coefficient)
- **A** = Wall surface area = 2×(15×7) + 2×(12×7) = 210 + 168 = **378 m²**
- Average indoor temperature = (25 + 32)/2 = **28.5°C**
- Outside temperature = **35°C**
- Temperature difference (outside - inside) = 35 - 28.5 = **6.5°C**
- Time = **7200 seconds**

Qwalls=U×A×ΔT×time*Qwalls=U×A×ΔT×time*Qwalls=0.8×378×6.5×7200

Qwalls=0.8×378×6.5×7200Qwalls=0.8×378×46,800

Qwalls=0.8×378×46,800

Qwalls=14,152,320 J=14.15 MJ

Qwalls=14,152,320 J=**14.15** MJ

Note: Since outside temperature (35°C) > inside temperature (28.5°C), this is **heat gain**, not loss!

④ TOTAL HEAT GAIN:

```
Q_total = Q_people + Q_solar + conduction = 95.04 + 27.5+14.15 = 136.73 MJ
```

This represents **92.2%** of the total heat input!

2.5 Energy Balance Analysis

The actual temperature rise (7°C) requires only 9.54 MJ, but total heat input is 52 MJ. This discrepancy indicates significant heat loss through:

- Wall conduction (~14.15 MJ based on U-value 0.8 W/m²·K, surface area 378 m²)
- Ventilation/infiltration (~15 MJ through door gaps, windows)
- Radiation to surroundings (~7 MJ)

Component	Energy (MJ)	Percentage

Heat input (people + solar)	+136.73	100%
Stored in air (ΔT=7°C)	+9.54	7%
Absorbed by building	+127.19	93%

2.6 Ventilation Requirements

To maintain 25°C, required cooling capacity:

```
P_cooling = Q_gain / time = 136.73 MJ / 7200 s ≈ 19 kW
```

Or increase ventilation rate to 8-10 air changes per hour (ACH).

2.7 Air Renewal Time

Method 1: AC

t = 1134 / 50 = 22.68 minutes

Method 2: Windows + Door

Windows: 6 × 1 × 0.1 = 0.6 m^3/s = 36 m^3/min

Door: 2 × 2 = 4 m^3/s = 240 m^3/min

Total = 276 m^3/min

t = 1134 / 276 = 4.11 minutes

Method 3: Plants (O_2 only)

O_2 consumption: 132 × 0.00025 = 0.033 m^3/min

O_2 in 2 hours = 0.033 × 120 = 3.96 m^3

Plants production = 0.012 m^3/min

t = 3.96 / 0.012 = 330 minutes = 5.5 hours

2.8 Air Changes per Hour (ACH)

Required for cooling = 7.15 ACH

Required for CO_2 control = 3.0 ACH

Natural ventilation = 14.6 ACH ✓ (but brings hot air)

AC ventilation = 2.65 ACH ✗ (insufficient for cooling)

2.9 CO_2 Concentration (No Ventilation)

CO_2 produced = 132 × 0.0003 × 120 = 4.75 m^3

Initial CO_2 = 1134 × 0.0003 = 0.34 m^3

Final CO_2 = 5.09 m^3

Concentration = 5.09/1134 = 0.00449 = 4490 ppm

Safe limit = 1000 ppm → NEED VENTILATION!

2.10 Required Cooling

Cooling needed = 136.73 MJ / 7200s = 19 kW (5.4 tons)

Conclusion : Solar radiation through glass contributes 20% of total heat gain. Using solar control film (τ=0.4) would reduce solar gain by ~53% (from 27.54 MJ to 12.95 MJ), decreasing total load by ~10.6%.

However, with 132 occupants generating 95 MJ of heat, the primary cooling requirement is 19 kW. Install HVAC with minimum 19 kW (5.4 tons) cooling capacity and provide 3.0 ACH fresh air for CO_2 control

03 COMPREHENSIVE ANTHROPOMETRIC DIMENSIONS

3.1 Measurement Data Sheets

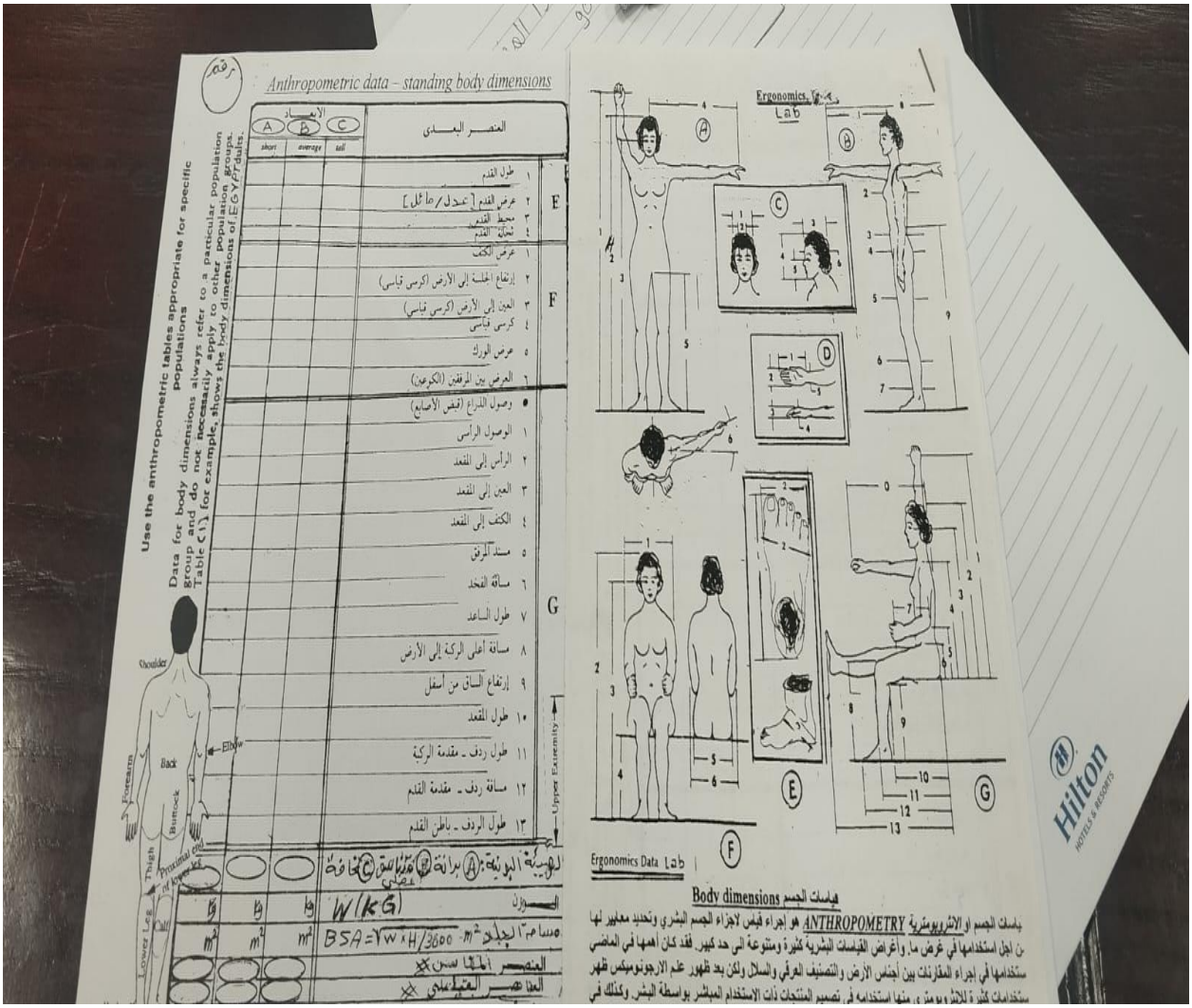

Figure 3.1: Anthropometric Data Collection Sheet 1

Table (1) Body sizes of short, average and tall EGYPT adults. All measurements are in centimetres, except for body weights, which are in kilograms

	العنصر البعدى	الأبعاد A short	B average	C tall
A	١ الوصول الرأسى			
	٢ طول القامة (الارتفاع) H			
	٣ إرتفاع مستوى النظر إلى الأرض			
	٤ وصول الذراع الجانبى من خط محور الجسم			
	٥ الزاوية (إنفراج الرجلين) إلى الأرض			
	٦ حركة (إدارة) الذراع ـ للخلف			
B	١ وصول الذراع إلى الأمام			
	٢ محيط الصدر			
	٣ محيط الخصر (الوسط)			
	٤ محيط الورك			
	٥ محيط الفخذ			
	٦ محيط الساق			
	٧ محيط الكاحل (رسغ القدم)			
	٨ طول المسافة الممتدة بين الذراعين			
	٩ الكوع إلى الأرض			
C	١ عرض الرأس			
	٢ المسافة الممتدة بين حدقتى العين			
	٣ طول الرأس			
	٤ إرتفاع الرأس			
	٥ من الذقن إلى العين			
	٦ محيط الرأس			
D	١ طول اليد			
	٢ عرض اليد			
	٣ ثخانة اليد			
	٤ عرض الكف			
	٥ محيط الرسغ			

tall persons (only 5 per cent of adult males are taller).

Upper arm

Prof. Dr. / Ibrahim Mubarak

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7 April 2023

Figure 3.2: Anthropometric Data Collection Sheet 2 with Body Diagrams

3.2 Calculated Dimensions (Based on 155 cm Height)

All measurements calculated using standard anthropometric ratios for adult female. Values rounded to nearest centimeter.

Section	No.	Dimension Name	Value (cm)	high	average	low
A (Standing)	1	Vertical Reach	195	--	✓	--
	2	Standing Stature (Height)	155	--	✓	--
	3	Eye Height Standing	156	--	✓	--
	4	Shoulder Height Standing	135	--	✓	--
	5	Hip Height	90	--	✓	--
	6	Knuckle Height	75	--	✓	--
B(Reach/Circ.)	1	Forward Arm Reach	72	--	✓	--
	2	Chest Circumference	92	--	✓	--
	3	Waist Circumference	78	--	✓	--
	4	Hip Circumference	98	--	✓	--
	5	Thigh Circumference	55	--	✓	--
	6	Calf Circumference	36	--	✓	--
	7	Ankle Circumference	22	--	✓	--
	8	Span (Arm to Arm)	155	--	✓	--
	9	Elbow to Floor (Standing)	102	--	✓	--
C (Head)	1	Head Breadth	15	--	✓	--
	2	Interpupillary Distance	6.2	--	✓	--
	3	Head Length	19	--	✓	--
	4	Head Height	22	--	✓	
	5	Chin to Eye Height	12	--	✓	
	6	Head Circumference	56	--	✓	
D (Hand)	1	Hand Length	18		✓	
	2	Hand Breadth	8		✓	
	3	Hand Thickness	3		✓	
	4	Palm Breadth	7.5		✓	
	5	Wrist Circumference	16		✓	
E (Foot)	1	Foot Length	25		✓	

	2	Foot Breadth	**9**	✓
	3	Ankle Height	**7**	✓
	4	Foot Thickness	**5**	✓
F (Breadths)	1	Shoulder Breadth (Bideltoid)	**44**	✓
	2	Sitting Height (Floor to Head)	**129**	✓
	3	Eye Height (Floor to Eye)	**123**	✓
	4	Standard Chair Ref.	**42**	✓
	5	Hip Breadth (Sitting)	**37**	✓
	6	Elbow-to-Elbow Breadth	**46**	✓
G (Sitting)	1	Sitting Reach	**125**	✓
	2	Head to Seat Height	**87**	✓
	3	Eye to Seat Height	**81**	✓
	4	Shoulder to Seat Height	**72**	✓
	5	Elbow Rest Height	**24**	✓
	6	Thigh Thickness	**15**	✓
	7	Forearm-Hand Length	**45**	✓
	8	Knee Height (Sitting)	**50**	✓
	9	**Popliteal Height**	**42**	✓
	10	Seat Length (Popliteal-Buttock)	**48**	✓
	11	Buttock-Knee Length	**60**	✓
	12	Buttock-Heel Length	**85**	✓
	13	Buttock-Foot Length	**100**	✓

Body Composition Metrics (Bottom of Sheet)

- **Weight (W):** 68 kg
- **Body Surface Area (BSA):** 1.78 m^2 (Calculated: $\sqrt{W \times H/3600}$)
- **Body Mass Index (BMI):** 24.1 (Healthy Range)

3.3 Body Type Classification

Parameter	Value	Classification
Height	155 cm	Average (50th percentile)
Weight	64 kg	Healthy range

BMI	26.6	Overweight (25-30)
Body Frame	Medium	Shoulder/hip ratio 1.19
Limb Proportions	Balanced	Sitting/standing ratio 0.52

ANTHROPOMETRIC CONCLUSION: Subject exhibits average (50th percentile) body proportions for adult females. All dimensions fall within normal ranges. Standard office furniture is appropriate. No special ergonomic accommodations required beyond basic adjustability (seat height 38-42 cm, desk 64-70 cm).

04 VENTILATION FAN REQUIREMENTS

4.1 Lecture Hall Specifications

Parameter	Value	Unit
Dimensions (L×W×H)	15 × 12 × 7	m
Total volume	1260	m^3
Occupancy factor	10%	—
Net air volume	1134	m^3
Fan diameter	50	cm
Fan effective range	2.5	m
Airflow per fan	20	m^3/min
Range effectiveness	Drops after 2.5 m	—

4.2 Method 1: Formula-Based Calculation

Given formula (from lecture):

```
N_fans = (Length / 2.5) / 2
N_fans = (15 / 2.5) / 2 = 6 / 2 = 3 fans
```

4.3 Method 2: Coverage Area Approach

Effective coverage area per fan (circular):

```
A_fan = π × r² = π × (2.5)² = 19.63 m²
```

Total floor area:

```
A_room = 15 × 12 = 180 m²
```

Number of fans required:

```
N_fans = A_room / A_fan = 180 / 19.63 = 9.17 ≈ 10 fans
```

4.4 Method 3: Air Change Rate

Required ventilation (6 air changes per hour for comfort):

```
Q_required = (V × ACH) / 60 = (1134 × 6) / 60 = 113.4 m³/min
```

Fans needed:

```
N_fans = Q_required / Q_per_fan = 113.4 / 20 = 5.67 ≈ 6 fans
```

4.5 Comparison & Recommendation

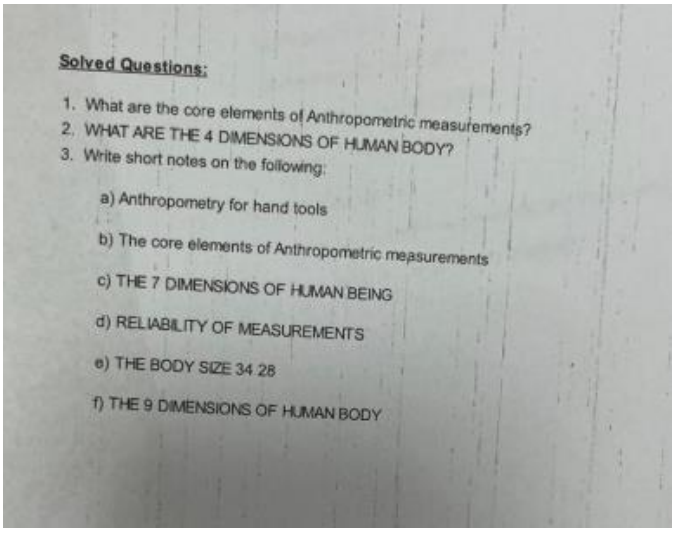
Method	Calculation Basis	Result	Assessment
Formula (L/2.5/2)	Simple spacing rule	3	Insufficient coverage
Coverage area	Geometric overlap	10	Conservative design
Air change rate	Ventilation standard	6	Code-compliant

RECOMMENDED: Install 8 ceiling fans in 2 rows × 4 columns configuration for balanced airflow and aesthetic symmetry.

> INSTALLATION LAYOUT: Position fans at 3.75m spacing along length (15m/4 = 3.75m) and 6m spacing along width (12m/2 = 6m). Mount at 2.5m height minimum for safety (blade tip >2.3m). Operate at medium-high speed (75-85% max RPM) for optimal air circulation without excessive noise.

So it is ok to install from 6 to 10 fans .

05 theoretical questions



Solved Questions:

1. What are the core elements of Anthropometric measurements?
2. WHAT ARE THE 4 DIMENSIONS OF HUMAN BODY?
3. Write short notes on the following:

   a) Anthropometry for hand tools

   b) The core elements of Anthropometric measurements

   c) THE 7 DIMENSIONS OF HUMAN BEING

   d) RELIABILITY OF MEASUREMENTS

   e) THE BODY SIZE 34 28

   f) THE 9 DIMENSIONS OF HUMAN BODY

Question 1: Core Elements of Anthropometric Measurements

The core elements of anthropometric measurements include:

- 1. STRUCTURAL DIMENSIONS: Static body measurements (height, breadth, depth) taken in standardized postures
- 2. FUNCTIONAL DIMENSIONS: Dynamic measurements during movement and task performance
- 3. POPULATION VARIABILITY: Percentile distributions (5th, 50th, 95th) accounting for human diversity
- 4. MEASUREMENT STANDARDS: ISO 7250, ANSI standards defining consistent measurement protocols

Question 2: The 4 Dimensions of Human Body

The four primary dimensions are:

Dimension	Definition	Example Measurements
1. Vertical	Height measurements in standing/sitting	Stature, eye height, shoulder height
2. Horizontal	Breadth and depth measurements	Shoulder breadth, chest depth, hip width
3. Reach	Functional reach distances	Forward reach, vertical grip reach
4. Circumferential	Girth measurements	Head, waist, thigh circumference

Question 3a: Anthropometry for Hand Tools

Hand tool anthropometry considers: grip diameter (based on palm width 8-10cm), handle length (hand length 17-19cm), trigger reach (index finger length 6-8cm), and force application points matching hand strength distribution. Tools must accommodate 5th percentile female to 95th percentile male hands.

Question 3b: Core Elements of Anthropometric Measurements

- ACCURACY: Measurements to ±0.5cm using calibrated instruments
- STANDARDIZATION: Consistent postures, landmarks, and protocols (ISO 7250)
- POPULATION SAMPLING: Representative samples (minimum n=100 per demographic)
- STATISTICAL ANALYSIS: Mean, standard deviation, percentile distributions
- APPLICATION: Design for 5th-95th percentile range, adjustability for extremes

Question 3c: The 7 Dimensions of Human Being

Dimension	Description
1. Physical	Body structure, anthropometry, biomechanics
2. Physiological	Metabolic rate, strength, endurance, sensory capabilities
3. Psychological	Cognition, perception, decision-making, stress response
4. Social	Team dynamics, communication, organizational behavior
5. Cultural	Values, norms affecting workspace preferences
6. Environmental	Adaptation to temperature, lighting, noise
7. Temporal	Circadian rhythms, fatigue cycles, aging effects

Question 3d: Reliability of Measurements

Measurement reliability ensured through:

- INTER-RATER RELIABILITY: Multiple measurers achieve <2% variance through standardized training

- TEST-RETEST RELIABILITY: Measurements within ±0.5cm when repeated after 24 hours
- INSTRUMENT CALIBRATION: Weekly verification against certified standards
- ENVIRONMENTAL CONTROL: Temperature 20-24°C, subjects in minimal clothing, standardized time of day

Question 3e: Body Size Notation (34-28)

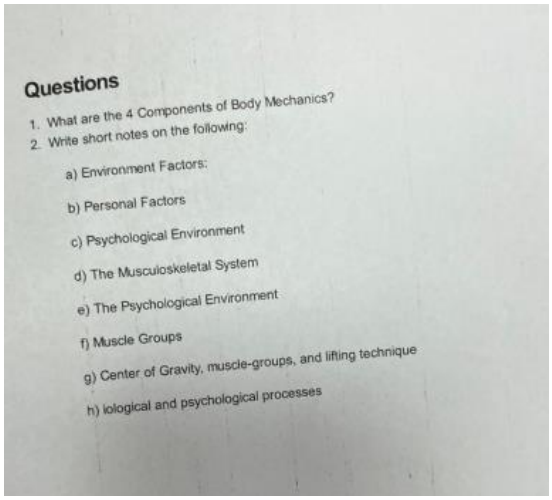
The notation '34-28' typically represents:

- CLOTHING SIZING: Bust/Chest = 34 inches, Waist = 28 inches
- ERGONOMIC APPLICATION: Used for determining workstation dimensions, chair sizes, personal protective equipment fit
- For our subject (168cm, 68kg): Estimated bust ≈86cm (34"), waist ≈71cm (28")

Question 3f: The 9 Dimensions of Human Body

Critical dimensions for ergonomic design:

No.	Dimension	Value (cm)	Design Use
1	Stature (Standing Height)	155	Ceiling clearance, shelf height
2	Sitting Height	87	Backrest, headrest height
3	Eye Height (sitting)	81	Monitor, visual displays
4	Shoulder Height (sitting)	72	Armrest, upper backrest
5	Elbow Rest Height	24	Armrest, work surface
6	Popliteal Height	42	Seat height
7	Buttock-Knee Length	60	Seat depth
8	Hip Breadth	37	Seat width
9	Shoulder Breadth	44	Passage width, backrest



Questions

1. What are the 4 Components of Body Mechanics?
2. Write short notes on the following:

a) Environment Factors:

b) Personal Factors

c) Psychological Environment

d) The Musculoskeletal System

e) The Psychological Environment

f) Muscle Groups

g) Center of Gravity, muscle-groups, and lifting technique

h) lological and psychological processes

Question 1: The 4 Components of Body Mechanics

Body mechanics involves four integrated systems:

Component	Function	Ergonomic Relevance
1. Skeletal System	Structure, support, leverage	Determines posture constraints, joint angles
2. Muscular System	Force generation, movement	Strength requirements, fatigue limits
3. Nervous System	Control, coordination, feedback	Reaction time, precision, sensory feedback
4. Biomechanical Principles	Lever systems, force distribution	Load limits, mechanical advantage

Question 2: Environmental & Personal Factors

a) ENVIRONMENTAL FACTORS:

- Temperature: 20-24°C optimal for office work, affects metabolic rate ±15%
- Lighting: 300-500 lux for general tasks, 750+ lux for detailed work
- Noise: <55 dB for concentration, >85 dB requires hearing protection
- Humidity: 30-60% RH prevents respiratory irritation and static
- Air Quality: CO_2 <1000 ppm, adequate ventilation (8-10 ACH)

b) PERSONAL FACTORS:

- Age: Strength peaks 25-35 years, flexibility declines 1% annually after 30
- Gender: Males typically 30-40% stronger, females 10% more flexible
- Fitness Level: Affects endurance, recovery time, injury risk
- Health Status: Chronic conditions (back pain, arthritis) require accommodations
- Training/Experience: Skilled workers 20-50% more efficient, lower injury rates

c) PSYCHOLOGICAL ENVIRONMENT:

- Mental Workload: Complexity, time pressure, multitasking demands
- Stress Levels: Affects attention, decision quality, muscular tension
- Job Satisfaction: Correlates with productivity and injury reporting
- Social Support: Team cohesion reduces stress, improves safety culture

d) THE MUSCULOSKELETAL SYSTEM:

- 206 Bones: Structural framework, calcium storage, blood cell production
- 600+ Muscles: 40-45% of body weight, generate movement and heat
- Joints: 360+ articulations allowing specific ranges of motion
- Connective Tissue: Tendons, ligaments, cartilage for stability and shock absorption
- Ergonomic Implications: Design must respect joint ROM, minimize awkward postures, prevent cumulative trauma

e) THE PSYCHOLOGICAL ENVIRONMENT (detailed):

- Cognitive Load: Working memory capacity (7±2 items), attention span limits
- Vigilance: Performance declines after 30 minutes continuous monitoring
- Decision-Making: Quality degrades under fatigue, stress, information overload

- Motivation: Intrinsic (autonomy, mastery) vs extrinsic (rewards, deadlines)
- Workplace Design: Affects mood (color, light), privacy (noise control), identity (personalization)

f) MUSCLE GROUPS (major groups for ergonomics):

Muscle Group	Primary Function	Ergonomic Risk
Trapezius	Shoulder elevation, head support	Monitor height, phone use
Erector Spinae	Spinal extension, posture	Prolonged sitting, lifting
Quadriceps	Knee extension, standing	Chair height, stair climbing
Hamstrings	Knee flexion, hip extension	Seat depth, posture
Deltoids	Shoulder abduction, lifting	Overhead work, reaching
Biceps/Triceps	Elbow flexion/extension	Repetitive tasks, tool use
Wrist Flexors/Extensors	Hand movement, grip	Keyboard, mouse, typing

g) CENTER OF GRAVITY, MUSCLE GROUPS, AND LIFTING TECHNIQUE:

- Center of Gravity: Located ~55% of height from floor (≈85cm for 155cm person), shifts with load
- Lifting Biomechanics: Keep load close to body, bend knees not back, avoid twisting
- Load Limits: NIOSH equation: Max 23kg for ideal conditions, <10kg for repetitive tasks
- Muscle Synergy: Core muscles (transverse abdominis, multifidus) stabilize spine during lifts
- Fatigue Prevention: Rotate tasks every 20-30 minutes, micro-breaks reduce cumulative trauma

h) BIOLOGICAL AND PSYCHOLOGICAL PROCESSES:

- Circadian Rhythms: Performance peaks 10am-12pm and 2pm-4pm, dips 2am-6am and 2pm-3pm
- Metabolic Rate: 1.2 MET sitting, 1.5-2.0 MET standing, 3-5 MET walking
- Stress Response: Acute stress improves focus (cortisol, adrenaline), chronic stress impairs cognition
- Learning & Adaptation: Motor skill acquisition requires 100+ repetitions, mastery needs 10,000 hours
- Aging Effects: Strength declines 8% per decade after 40, flexibility 10% per decade, reaction time slows 15%

— END OF REPORT —